# Cumulative Active-Clause Entropy as a Computable Proxy for Resolution Cumulative Space: An Honest Empirical Localisation on Tseitin and Structured Families

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**Abstract**

We study a computable, per-trace functional $\Phi_{\text{count}}(\pi) := \sum_t |M_t|$ — the sum over a Davis–Putnam refutation trace of the active-clause-set size at each step — and its relationship to the formula-level invariant $\text{CSpace}_{\text{cum}}(F)$ of Alwen, de Rezende, Nordström and Vinyals (ITCS 2017, paper 38; henceforth ADRNV) [1]. Our contribution is **an honest empirical localisation**, *not* a complexity-theoretic separation. We (i) state Lemma A in its honest two-part form — a near-tautological per-trace identity discharged by definitional unfolding in Lean 4, plus a corrected formula-level **schedule-bookkeeping inequality** $\Phi^{\text{DP},h}_{\text{count}}(F) \leq \text{CSpace}_{\text{cum}}(\pi'_{\text{DP},h})$ via a primitive expansion of the DP macro-trace — and **retract** the v4 reverse direction $\Phi_{\text{count}} \geq \text{CSpace}_{\text{cum}}$, which v4 derived from an erroneous one-line min-over-traces argument; (ii) report a pre-registered measurement campaign with cluster-robust model selection and paired-by-seed bootstrap. The combined DP-min-occurrence corpus on rand-3-regular Tseitin over $n \in \{14, 22, 30, 32, 34, 36\}$ gives a log-log slope $\widehat{\alpha}_\Phi = 4.43$, **up from the v3 figure of 3.43 on** $n = 10$–$30$ and continuing to drift upward across windows ($2.42 \to 2.93 \to 3.43 \to 4.43$); the $n = 36$ point is left-censored (3 of 10 instances `BLEW_UP` at MAX_DB $= 1.5 \times 10^6$), biasing the slope **downward**, with a Tobit correction flagged as future work. Per the corrected Lemma A, this exponent is reported as a **schedule-specific measurement with no formal ordering to** $\text{CSpace}_{\text{cum}}$, not as an upper witness. (iii) The DRAT/DP gap is confirmed on paired-by-seed instances: $\widehat{\alpha}_{\text{DRAT,default}} - \widehat{\alpha}_{\text{DP,min-occ}} \approx 3.28 \pm 0.4$ on $n \in \{14, 22, 30\}$. (iv) The v3 mechanism narrative for that gap — "inprocessing eliminates clauses, hence lower DRAT $\Phi$" — is **retracted**. A flag-by-flag decomposition on $n \in \{10, \ldots, 60\}$ with a paired-by-seed cluster bootstrap (B=2000) shows the no-restart effect *survives* with full-range CI $[+0.213, +0.435]$ but is fragile: excluding the $n = 60$ cell collapses the shift to $+0.07$ $[+0.03, +0.11]$. No-inprocessing is reported as suggestive ($+0.14$ $[+0.07, +0.18]$ full range; CI crosses zero without $n = 60$). No-eliminate is null. Roughly 10% of the DRAT/DP exponent gap is attributable to restart strategy, with the caveat that the signal is concentrated at $n = 60$. (v) On $Q_5$ Tseitin, DP-min-occurrence deterministically aborts at step 37 with final database $2^{20}$ at MAX_DB $= 4 \times 10^6$ across all three seeds; the $Q_5$ slope is therefore **not measurable** under this heuristic, and the v3 "treewidth not degree" framing is reframed as a structural negative. (vi) On random-4-regular Tseitin (DP-infeasible: 5/5 `BLEW_UP` at $n = 16$), kissat-DRAT gives slope 7.69 on $n \in \{16, 20, 24, 28\}$; this closes the v3 symmetric-comparison defect at the cost of a proof-system confound (DP on $Q_4$ vs DRAT on rand_4reg), which we acknowledge rather than wave away. The headline reading of this paper is: **a cumulative-entropy proxy $\Phi$ for resolution refutations on Tseitin formulas, with a measured exponent $\widehat{\alpha}_\Phi \approx 4.43$ reported as a heuristic-specific quantity with no formal ordering to cumulative space, after surgical retractions of the v4 Lemma A direction and the $\Omega(n^2)$ Corollary C2 chain.** The exercise is offered as a methodological template for cumulative-space empirics, not as evidence bearing on P vs NP. The ADRNV open problem on Tseitin (p. 38:19, lines 996–998) is reaffirmed as open and is **not** addressed.

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# Contents

# 1 Introduction

Cumulative clause-space [1] is the sum, along a resolution-with-memory trace $\pi$, of the sizes of the active memory configurations $M_t$. ADRNV define it formula-level as a minimum over admissible traces, and pose as an open problem (p. 38:19, lines 996–998) the extension of quadratic cumulative-space lower bounds beyond pebbling formulas — in particular to Tseitin.

We do not address that open problem. We study a strictly more accessible quantity: for a deterministic Davis–Putnam refutation [5] with the **minimum-occurrence elimination heuristic** (DP, min-occ), record at each elimination step $t$ the current active clause set $M_t$ and report

$$\Phi_{\text{count}}(\pi) := \sum_t |M_t|.$$

$\Phi_{\text{count}}$ is per-trace, fixed-heuristic, and computable in pure stdlib. The relationship to $\text{CSpace}_{\text{cum}}$ is delicate, and Lemma A (§3) makes it precise:

- *Per-trace Lean identity*: for the list-of-`ProofState`s representation $\pi$ we measure, the Lean function `cumulativeEntropy` is definitionally equal to $\sum_t |\sigma_t.\text{activeClauses}|$. This is a near-tautology of the Lean definitions; **no ADRNV semantics are mechanised by it**.
- *Formula-level schedule-bookkeeping inequality (corrected in v5)*: $\Phi_{\text{count}}(\text{DP}, \text{min-occ})(F) \leq \text{CSpace}_{\text{cum}}(\pi'_{\text{DP},h})$ where $\pi'_{\text{DP},h}$ is the canonical primitive (download / inference / erasure) expansion of the DP macro-trace. The v4 directional claim $\Phi_{\text{count}} \geq \text{CSpace}_{\text{cum}}(F)$ is **retracted** (see §3 and §3.5): DP elimination steps are macros, not primitive res-with-memory operations, and the macro-checkpoint sum does not majorise the formula-level min-over-traces. The identification of the Lean `ProofState.activeClauses` with the ADRNV memory configuration is **informal pending a Lean port of ADRNV Definition 11**; we make this explicit in the Lean stub as a labelled `sorry`.

Hence every exponent we report is a **schedule-specific measurement** of $\Phi_{\text{count}}(\text{DP}, \text{min-occ})$, with **no formal ordering** to $\text{CSpace}_{\text{cum}}$. This paper is the v5 draft. It explicitly retracts portions of v1, v2, v3, and v4 (§3.5), and incorporates the v3.5 surgical fixes (Lemma A honesty, SC3 7/0/1 sign-test recount, full censoring report).

## 1.1 What this paper is, and is not

This paper is an empirical methodology contribution: a pre-registered measurement protocol with cluster-robust statistics, three-model selection, censoring-aware fits, paired-by-seed bootstrap, and a transparent retraction trail. It is *not* a complexity-theoretic result. We do not claim a separation, a new lower bound, or a tightening of any theorem in ADRNV. Section 3 explains in what restricted formal sense the empirical work could ever bear on cumulative-space — and Lemma A delimits that sense precisely.

# 2 Definitions

## 2.1 The Lean anchor (verbatim from `Conjecture003.lean` lines 369–411)

```
structure ProofState (V : Type) [DecidableEq V] where
  activeClauses : Finset (Clause (Sym2 V))
  step : Nat

noncomputable def proofStateEntropy (sigma : ProofState V) : Nat :=
  sigma.activeClauses.card

noncomputable def cumulativeEntropy (states : List (ProofState V)) : Nat :=
```

```
  states.map proofStateEntropy |>.sum

noncomputable def totalLiteralWeight (sigma : ProofState V) : Nat :=
  sigma.activeClauses.sum Finset.card
```

`activeClauses` is a `Finset` (no multiplicities); `step` is a pure index. `cumulativeEntropy` is the sum over a `List ProofState` of `proofStateEntropy`. This is the Lean-side definition we point at.

### 2.2 ADRNV side

ADRNV [1] (Definition 11, paper 38) define a memory configuration $M_t$ along a resolution-with-memory trace; the **per-trace** cumulative-space is

$$\mathrm{CSpace}_{\mathrm{cum}}(\pi) := \sum_t |M_t|,$$

and the **formula-level** invariant is

$$\mathrm{CSpace}_{\mathrm{cum}}(F) := \min_{\pi \text{ admissible}} \mathrm{CSpace}_{\mathrm{cum}}(\pi).$$

Footnote 1 (p. 38:3) confirms the $\sum_t |M_t|$ reading.

### 2.3 The measurement $\Phi_{\mathrm{count}}(\pi)$ we report

For a fixed DP refutation with min-occurrence elimination, we record the sequence of `ProofStates` $\sigma_0, \sigma_1, \ldots, \sigma_T$ and define

$$\Phi_{\mathrm{count}}(\pi) := \sum_{t=0}^{T} |\mathrm{activeClauses}(\sigma_t)|.$$

Numerically `Phi_count(`$\pi$`) = cumulativeEntropy(states)` on the Lean side.

### 2.4 The load-bearing distinction

| Object | Type | Quantifier |
|---|---|---|
| $\Phi_{\mathrm{count}}(\pi)$ | per-trace, our measurement | none (fixed $\pi$) |
| $\mathrm{CSpace}_{\mathrm{cum}}(\pi)$ | per-trace, ADRNV's Def 11 | none (fixed $\pi$) |
| $\mathrm{CSpace}_{\mathrm{cum}}(F)$ | formula-level, ADRNV's invariant | **min over** $\pi$ |
| $\Phi_{\mathrm{count}}(\mathrm{DP}, \text{min-occ})(F)$ | formula-level under heuristic | DP min-occ fixes $\pi$ |

The v5 corrected relation is the schedule-bookkeeping inequality $\Phi^{\mathrm{DP},h}_{\mathrm{count}}(F) \le \mathrm{CSpace}_{\mathrm{cum}}(\pi'_{\mathrm{DP},h})$, with **no** formal ordering between $\Phi_{\mathrm{count}}(\mathrm{DP}, \text{min-occ})(F)$ and the formula-level $\mathrm{CSpace}_{\mathrm{cum}}(F)$. The v1/v2 conflation and the v4 reverse direction are retracted in §3.5.

## 3 Lemma A (honest two-part statement, v5 corrected)

**Lemma 1** (part i — per-trace identity, near-tautology)**.** *For any list of `ProofState V` instances $\pi = (\sigma_0, \ldots, \sigma_T)$, the Lean function `cumulativeEntropy` from §2.1 satisfies*

$$\mathtt{cumulativeEntropy}(\sigma_0, \ldots, \sigma_T) \;=\; \sum_{t=0}^{T} |\sigma_t.\mathtt{activeClauses}|$$

*by definitional unfolding of the two Lean definitions:*

```
cumulativeEntropy = (List.map proofStateEntropy) |>.sum
proofStateEntropy sigma := sigma.activeClauses.card
```

***No ADRNV semantics are mechanised by this statement.*** *It is a near-tautology of the Lean definitions and is included for record-keeping, not for content.*

**Lemma 2** (part ii — formula-level relation, schedule-bookkeeping inequality, v5 corrected)**.** *We replace the v4 draft's "immediate from min-over-traces" wording, which conflated coarse-grained DP elimination steps with primitive resolution-with-memory steps in the sense of ADRNV Definition 11.*

*Setup.* ADRNV's resolution-with-memory model [1] has three primitive operations: (a) **download** a clause of $F$ into the memory configuration $M$, (b) **inference**, which adjoins to $M$ a resolvent of two clauses already in $M$, (c) **erasure**, which deletes a clause from $M$. The cumulative space of an admissible trace $\pi' = (M_0, M_1, \ldots, M_{T'})$ is $\mathrm{CSpace}_{\mathrm{cum}}(\pi') := \sum_{t=0}^{T'} |M_t|$, and $\mathrm{CSpace}_{\mathrm{cum}}(F) := \min_{\pi' \text{ admissible refuting } F} \mathrm{CSpace}_{\mathrm{cum}}(\pi')$.

A Davis–Putnam elimination step on variable $v$ is **not** a primitive res-with-memory operation. It is a macro: download every clause of the current active set that mentions $v$ (if not yet downloaded), form every pairwise resolvent on $v$, and erase every clause mentioning $v$. The Lean `ProofState.activeClauses` after the $t$-th elimination corresponds to a memory configuration $M_t^{\mathrm{DP}}$, but $\Phi^{\mathrm{DP},h}_{\mathrm{count}}(F) := \sum_t |M_t^{\mathrm{DP}}|$ sums only over **macro-checkpoints**, one per eliminated variable.

*Re-scheduling claim.* Any DP-min-occ run on $F$ can be expanded into an admissible res-with-memory trace $\pi'$ whose checkpoint memory states at the macro-boundaries coincide with the $M_t^{\mathrm{DP}}$. Concretely, between checkpoints $t$ and $t+1$ the expansion inserts (i) the downloads of the not-yet-downloaded $v_{t+1}$-clauses, (ii) the pairwise resolutions on $v_{t+1}$, (iii) the erasures of the original $v_{t+1}$-clauses. Each intermediate $|M|$ in this expansion is **at least** the smaller of $|M_t^{\mathrm{DP}}|$ and $|M_{t+1}^{\mathrm{DP}}|$ (monotone phases) and may transiently exceed both during the resolution phase before the matching erasures.

*Consequence — the honest direction.* Because $\pi'$ contains additional summands (the primitive intermediates), $\mathrm{CSpace}_{\mathrm{cum}}(\pi') \geq \Phi^{\mathrm{DP},h}_{\mathrm{count}}(F)$. Taking the min over all admissible refutations yields

$$\mathrm{CSpace}_{\mathrm{cum}}(F) \;\leq\; \mathrm{CSpace}_{\mathrm{cum}}(\pi') \;\not\leq\; \Phi^{\mathrm{DP},h}_{\mathrm{count}}(F).$$

The min-over-traces argument therefore does **not** discharge $\Phi^{\mathrm{DP},h}_{\mathrm{count}}(F) \geq \mathrm{CSpace}_{\mathrm{cum}}(F)$. The v4 draft's one-line justification ("immediate from min-over-traces: DP under any fixed $h$ realises one specific admissible trace") is wrong: the DP trace is not, qua sum, a res-with-memory trace, and its primitive expansion has strictly more summands than $\Phi_{\mathrm{count}}$ counts.

*Restated formula-level relation (corrected, v5).* The strongest defensible statement is the **weaker** schedule-specific one:

$$\Phi^{\mathrm{DP},h}_{\mathrm{count}}(F) \;=\; \sum_{t=0}^{T} |M_t^{\mathrm{DP},h}| \;\leq\; \mathrm{CSpace}_{\mathrm{cum}}(\pi'_{\mathrm{DP},h}),$$

where $\pi'_{\mathrm{DP},h}$ is the canonical primitive expansion above. We do **not** claim $\Phi^{\mathrm{DP},h}_{\mathrm{count}}(F) \geq \mathrm{CSpace}_{\mathrm{cum}}(F)$. We claim only that the macro-checkpoint sum $\Phi_{\mathrm{count}}$ is a **lower estimator** of the cumulative space of *one* admissible refutation (its own expansion), and that this expansion is one of the traces over which $\mathrm{CSpace}_{\mathrm{cum}}(F)$ takes its minimum. The directional consequence for the empirical work in §7 is:

$$\Phi^{\mathrm{DP},h}_{\mathrm{count}}(F) \;\leq\; \mathrm{CSpace}_{\mathrm{cum}}(\pi'_{\mathrm{DP},h}), \qquad \mathrm{CSpace}_{\mathrm{cum}}(F) \;\leq\; \mathrm{CSpace}_{\mathrm{cum}}(\pi'_{\mathrm{DP},h}).$$

Both quantities are lower-bounded by the same admissible-trace cumulative space, but they are **not** ordered with respect to one another by this argument.

*Honest cascade into Corollary C2.* The v4 Corollary C2 chain

$$\Phi^{\mathrm{DP,min\text{-}occ}}_{\mathrm{count}}(F_n) \;\geq\; \mathrm{CSpace}_{\mathrm{cum}}(F_n) \;\geq\; c \cdot n^2$$

no longer goes through unconditionally. The second inequality is downgraded separately in §9 (E2) to a linear bound; the first does **not** follow from Lemma A as corrected here. C2 should therefore be downgraded on two grounds: $\mathrm{CSpace}_{\mathrm{cum}}(F_n) = \Omega(n)$ stands as a literature-derived lower bound (the $\Omega(n^2)$ does not), and $\Phi^{\mathrm{DP,min\text{-}occ}}_{\mathrm{count}}(F_n)$ remains a measured exponent on a fixed heuristic with **no formal ordering** to $\mathrm{CSpace}_{\mathrm{cum}}(F_n)$. The empirical $\alpha_\Phi \approx 4.43$ being **larger** than 2 is consistent with $\Phi_{\mathrm{count}}$ sitting above $\mathrm{CSpace}_{\mathrm{cum}}$ on this regime, but it is not a proof of the ordering.

**Honesty disclosure.** Path A (showing the DP-trace sum *equals* a primitive res-with-memory cumulative space) does not close the gap: expansion strictly adds summands, so the expansion's cumulative space is $\geq \Phi_{\mathrm{count}}$, not $=$. The v4 directional claim $\Phi_{\mathrm{count}} \geq \mathrm{CSpace}_{\mathrm{cum}}$ is **retracted**. The "upper witness" language in the v4 abstract (line 11) and §7 — which depended on the retracted direction — is replaced throughout v5 with "schedule-specific measurement with no formal ordering to $\mathrm{CSpace}_{\mathrm{cum}}$".

**Lean stubs (v5: reflect the corrected one-sided bound, no false reverse direction).**

```
/-- (i) Near-tautology of the Lean definitions; mechanises only the
    internal identity, *not* the bridge to ADRNV semantics. -/
theorem cumulativeEntropy_unfold
    {V : Type} [DecidableEq V] (states : List (ProofState V)) :
    cumulativeEntropy states =
      (states.map (.activeClauses.card)).sum := by
  unfold cumulativeEntropy proofStateEntropy
  rfl -- definitional

/-- (ii) The ADRNV bridge, v5 corrected. The statement is the WEAKER
    direction: Phi_count(DP,h)(F) <= CSpace_cum(pi'_{DP,h}) where
    pi' is the canonical primitive expansion of the DP macro-trace.
    The proof is a `sorry` until a Lean port of ADRNV Definition 11
    exists and the primitive expansion is mechanised.
    THIS IS A DOCUMENTED DEFINITIONAL BRIDGE, NOT A MISSING PROOF STEP. -/
theorem phi_count_le_CSpace_cum_pi_prime
    {V : Type} [DecidableEq V] (pi : DP_trace V) :
    Phi_count pi <= CSpace_cum_ADRNV (primitive_expansion pi) := by
  sorry -- pending Lean port of ADRNV Def 11 + expansion construction
```

**Verbatim disclosure (this paragraph must appear in any republication).**

> Lemma A (i) is a near-tautology of Lean's definition; the bridge to ADRNV semantics in (ii) is informal pending a Lean port of ADRNV Definition 11. We do not claim a mechanised proof of (ii). The v5-corrected direction is the **weaker** $\Phi^{\mathrm{DP},h}_{\mathrm{count}}(F) \leq \mathrm{CSpace}_{\mathrm{cum}}(\pi'_{\mathrm{DP},h})$; the v4 reverse direction $\Phi_{\mathrm{count}} \geq \mathrm{CSpace}_{\mathrm{cum}}(F)$ is retracted. The chain in v4 Corollary C2 ($\Phi_{\mathrm{count}} \geq \mathrm{CSpace}_{\mathrm{cum}} \geq \Omega(n^2)$ on bounded-degree expander Tseitin) is **withdrawn** in v5 and replaced by the linear bound $\mathrm{CSpace}_{\mathrm{cum}}(\mathrm{Tseitin}_n) = \Omega(n)$ via Esteban–Torán [2] alone (§9).

**Corollary 1** (consequences for the empirical claims of this paper, v5 corrected)**.** *(C1, restated) Every empirical exponent we report for* $\Phi_{\mathrm{count}}(\mathrm{DP}, \text{min-occ})$ *is a* ***schedule-specific measurement*** *with no formal ordering to the formula-level* $\mathrm{CSpace}_{\mathrm{cum}}(F)$*. The v4 "upper witness" language is withdrawn; we do* not *claim that* $\Phi_{\mathrm{count}}(\mathrm{DP}, \text{min-occ})(F_n) = \Theta(n^\alpha)$ *implies* $\mathrm{CSpace}_{\mathrm{cum}}(F_n) = O(n^\alpha)$ *or any other directional bound on* $\mathrm{CSpace}_{\mathrm{cum}}$*. What survives is the*

*one-sided* $\Phi^{\mathrm{DP},h}_{\mathrm{count}}(F) \le \mathrm{CSpace}_{\mathrm{cum}}(\pi'_{\mathrm{DP},h})$*, where* $\pi'_{\mathrm{DP},h}$ *is the canonical primitive expansion of the DP macro-trace; this expansion is one admissible refutation, not the min-achieving one in general.*
*(C2, v5 revised — see §9 for full text) On bounded-degree expander Tseitin formulas* $\{F_n\}$*, Esteban–Torán [2] gives* clause-space$(F_n) \ge n - O(1)$*, hence per-trace* $\mathrm{CSpace}_{\mathrm{cum}}(\pi) \ge \text{max–space}(\pi) \ge n - O(1)$ *and therefore* $\mathrm{CSpace}_{\mathrm{cum}}(\mathrm{Tseitin}_n) = \Omega(n)$*. The v4 chain through ADRNV Lemma 12 to* $\Omega(n^2)$ *does* ***not*** *go through: Lemma 12's* $s^2$ *growth requires the refutation to spend* $\Omega(s)$ *steps at space* $\Omega(s)$*, while Esteban–Torán only guarantees the* peak *is* $\Omega(n)$*, not a sustained* plateau. *Extending the* $\Omega(n^2)$ *bound from ADRNV's XOR-pebbling formulas (Thm 14–15) to Tseitin is the open problem stated by ADRNV [1] (p. 38:19 lines 996–998), and remains open. The measured* $\Phi$*-exponent of §7 has* ***no formal relation*** *to either bound under v5-corrected Lemma A.*

**Caveat.** All exponents above 2 that we report on Tseitin-like families are *heuristic-conditional and schedule-conditional*: they are exponents of $\Phi_{\mathrm{count}}(\mathrm{DP}, \text{min-occ})$, not of $\mathrm{CSpace}_{\mathrm{cum}}$, and (after v5) carry no formal directional inequality to the formula-level cumulative space. The lack of an ordering between $\Phi_{\mathrm{count}}(\mathrm{DP}, \text{min-occ})(F)$ and $\mathrm{CSpace}_{\mathrm{cum}}(F)$ is the central methodological hazard of this paper and is not closed empirically.

This closes panel concerns **R1** (Lemma A was over-stated as an identity), **R7** (Lemma A asserted but not inline-proved), **v3-D1** (the Lean stub was a tautology, not a bridge), and the v5 panel concern **E1** (v4's reverse direction $\Phi_{\mathrm{count}} \ge \mathrm{CSpace}_{\mathrm{cum}}$ was unjustified).

### 3.5 Retraction block (v1, v2, v3, v4 claims withdrawn)

We list, in one place, exactly what is withdrawn from earlier drafts of this paper.

**Retracted from v1 (2026-05-15) and v2 (2026-05-24):**

1. **The single $n^{2.93}$ headline exponent.** v1 and v2 reported a single power-law fit on the DP corpus. The slope continues to drift across n-windows: 2.42 ($n \le 16$) → 2.93 ($n \le 28$) → 3.43 ($n \le 30$, v3 headline) → **4.43** ($n \le 36$, v4 combined paired + push-n). There is no observed asymptotic stabilisation. The single-exponent v1/v2 headline is retracted, the v3 figure is updated to 4.43, and even 4.43 is reported with the explicit drift caveat and an n=36 left-censoring acknowledgment.

2. **SC2 (order invariance) "supported".** v2 §3 table marked SC2 supported. The pre-registration registry verbatim text (§5) and the audit of paired cross-order bootstrap (§6) yield SC2 = **refuted**: order matters at the family level.

3. **SC4 (prover invariance) "supported".** Likewise retracted to **refuted**: paired-by-seed DP vs DRAT-default bootstrap gives $\widehat{\alpha}_{\text{DP,min-occ}} = 4.26$ vs $\widehat{\alpha}_{\text{DRAT,default}} = 7.54$ on the v4 paired corpus at $n \in \{14, 22, 30\}$ (§7), incompatible with prover-invariant scaling.

4. **The over-stated Lemma A identity.** v2 stated Lemma A as the equality $\Phi_{\mathrm{count}} = \mathrm{CSpace}_{\mathrm{cum}}$ at formula level. This is wrong: $\mathrm{CSpace}_{\mathrm{cum}}$ is min-over-traces, $\Phi_{\mathrm{count}}$ is fixed-trace. The corrected statement is the per-trace identity plus the v5 formula-level schedule-bookkeeping inequality of §3 above.

5. **"v2 BLEW_UP at grid_2D $n = 36, 49$" attributed to instance hardness.** The v2 abort points were a **harness artifact**, not an instance-level fact (re-run with the optimised DP finishes grid_2D $n = 25$ in 0.23 s). Treewidth-headline claims that leaned on the small-n window are demoted to "consistent with" rather than "establishes" until the larger-n grid runs (§12) clear the panel R3 concern.

**Retracted from v3 (2026-05-30):**

6. **The v3 §10 mechanism narrative for the DRAT/DP gap.** v3 attributed the gap to kissat inprocessing on the basis of a single-flag test with `-no-inprocessing` at small n. The B.2 flag-by-flag experiment at $n \in \{10, \ldots, 60\}$ (§10) **falsifies** the inprocessing-as-mechanism claim: at $n \leq 26$ all four flag settings give *byte-identical* $\Phi$ (inprocessing simply does not fire on these instances at small n); at $n \geq 30$ the dominant flag-controlled contribution is the **restart strategy**, not inprocessing or elimination. Disabling restarts (`-restart=false`) adds +0.323 to the DRAT log-log slope on $n \in \{10, \ldots, 60\}$ (default 6.688 $\to$ no-restart 7.011); v5 §10 confirms the effect survives a B=2000 cluster-by-seed paired bootstrap on the full 9-point fit but flags its concentration at $n = 60$. The remaining $\sim$90% of the $\sim$3.3-in-exponent DRAT/DP gap is intrinsic to the CDCL trace structure and is **not** localised to any single inprocessing stage. The v3 §10 mechanism is retracted; the corrected mechanism appears in §10 below.

7. **The v3 §7 $Q_4$ vs rand_4reg subsection.** v3 reported a "treewidth not degree" headline based on $Q_4$ under DP (slope $\approx 6.06$) without a rand_4reg comparator at all (rand_4reg was 5/5 `BLEW_UP` under DP at $n = 16$, omitted from v3 §7 — this is the v3.5 §C defect). v4 reports rand_4reg under **kissat-DRAT** at $n \in \{16, 20, 24, 28\}$, slope **7.69** (§7.3); and $Q_5$ under DP at $n = 32$ as a deterministic structural negative — abort at step 37 with final DB $2^{20}$ at MAX_DB $= 4 \times 10^6$ across all three seeds (§7.4). The v3 headline is retracted; the comparison is reframed as one across two proof systems (DP for $Q_4$, DRAT for rand_4reg) with the proof-system confound explicit.

**Retracted/downgraded from v4 (2026-06-01):**

8. **The v4 Lemma A (part ii) directional claim $\Phi^{\mathrm{DP},h}_{\mathrm{count}}(F) \geq \mathrm{CSpace}_{\mathrm{cum}}(F)$ is withdrawn.** The DP elimination trace is **not**, qua sum, a primitive res-with-memory trace; its canonical primitive expansion has strictly more summands than $\Phi_{\mathrm{count}}$ counts, yielding only the one-sided inequality $\Phi^{\mathrm{DP},h}_{\mathrm{count}}(F) \leq \mathrm{CSpace}_{\mathrm{cum}}(\pi'_{\mathrm{DP},h})$. The v4 one-line justification ("immediate from min-over-traces") was wrong. See the corrected statement in §3 above.

9. **The v4 Corollary C2 quadratic $\Omega(n^2)$ lower bound on $\mathrm{CSpace}_{\mathrm{cum}}(\mathrm{Tseitin}_n)$ via ADRNV Lemma 12 + Esteban–Torán is withdrawn.** Esteban–Torán guarantees the *peak* clause-space on Tseitin expanders is $\Omega(n)$, not that the peak is sustained as a plateau; ADRNV Lemma 12's $s^2$ growth requires the refutation to spend $\Omega(s)$ steps at space $\Omega(s)$, and the literature does not deliver that plateau for Tseitin. The defensible lower bound via this route is the linear $\mathrm{CSpace}_{\mathrm{cum}}(\mathrm{Tseitin}_n) \geq \mathrm{CSpace}(\mathrm{Tseitin}_n) \geq n - O(1) = \Omega(n)$. Extending to $\Omega(n^2)$ on Tseitin is the ADRNV open problem [1] (p. 38:19 lines 996–998) and is **not** delivered by this paper. See §9 below.

10. **The v4 "upper witness" language in the abstract and §7 (which depended on the retracted Lemma A direction) is withdrawn** and replaced throughout v5 with "schedule-specific measurement with no formal ordering to $\mathrm{CSpace}_{\mathrm{cum}}$".

11. **The v4 "+0.323 no-restart" headline is not retracted but downgraded.** The B=2000 cluster-by-seed paired bootstrap on the full 9-point fit gives CI $[+0.213, +0.435]$ (excludes zero), but excluding the $n = 60$ cell collapses the shift to +0.073 with CI $[+0.031, +0.111]$ (still excludes zero but 4× smaller in magnitude). No-inprocessing is downgraded from "negligible" to "suggestive" (+0.137 full-range, CI $[+0.070, +0.179]$; without $n = 60$ the CI $[-0.048, +0.316]$ crosses zero). No-eliminate is null (+0.030 full-range, CI $[-0.005, +0.073]$). See §10 below.

This block closes panel concern **R4** (no explicit retraction paragraph), v3 defects **D5**, **D7**, **D10**, and the v5 panel concerns **E1**, **E2**, **E3**. SC swap/drift details: panel concerns **N4** and **R5**.

## 4 Related work

ADRNV [1] introduce cumulative clause-space and prove quadratic lower bounds on pebbling-type formulas, leaving Tseitin as an open problem (p. 38:19). Esteban–Torán [2] give the linear clause-space lower bound for Tseitin on expanders. The Nordström space-width trade-off [3] and the line of Urquhart [4] hard instances are the closest theoretical antecedents. Solver-level measurement of resolution-resource proxies (Järvisalo–Heule–Biere, Elffers et al.) are the closest empirical antecedents — but to our knowledge no prior work has measured $\Phi_{\text{count}}(\text{DP}, \text{min-occ})$ systematically with the pre-registered statistical apparatus we use here. We do **not** claim to outperform these works empirically; we offer a complementary, hedged-by-design measurement on a different functional ($\Phi$ vs solver-step proxies). Closing panel concern **R10** (side-by-side empirical comparison with Järvisalo–Heule–Biere or Elffers et al.) is a stated **open task** (§12); the present draft does not include such a head-to-head.

## 5 Pre-registration (SC1–SC6, verbatim from the registry)

The success criteria below are reproduced **verbatim** from the project registry (workflow log b82c4fb62). They were registered prior to the v2 measurement campaign and have not been edited since.

| # | Verbatim SC text (registry) | STATUS in v5 | Adjudication |
|---|---|---|---|
| SC1 | *"$\Phi_{\text{count}}$ grows super-linearly in $n$ on at least one Tseitin-like expander family, with a bootstrap slope CI excluding 1.0."* | partial (drift) | §7.1 |
| SC2 | *"$\Phi_{\text{count}}$ is invariant in family-level scaling under permutation of clause input order, paired by seed."* | refuted (swap) | §6.3 |
| SC3 | *"On a degree-matched random-3-regular baseline, the Tseitin family has a strictly higher $\Phi$-level at matched $n$ in a sign-test sense."* | supported (aligned) | §7.2 |
| SC4 | *"The $\Phi$ exponent is invariant within a multiplicative factor under change of refutation back-end (DP vs DRAT)."* | refuted (downgraded in v5) | §10 |
| SC5 | *"The selected single-model fit (power, exponential, or stretched-exp) is stable to the choice of $n$-grid window."* | partial (drift) | §7.5 |
| SC6 | *"Lemma A (per-trace identity $\Phi_{\text{count}} = \text{CSpace}_{\text{cum}}$) holds; the formula-level monotonicity $\Phi_{\text{count}}(\text{DP}, \text{min-occ})(F) \geq \text{CSpace}_{\text{cum}}(F)$ is stated correctly."* | partially refuted in v5 | §3 |

**SC1 detail:** super-linearity met; the $\Omega(n^2)$ lower bound from v4 C2 is **withdrawn** in v5 and replaced by $\Omega(n)$; the *specific* fit-window-stable exponent claim drifts from the registry wording.

**SC4 detail:** restart strategy contributes ~10% of the exponent gap on the full-range fit but the signal concentrates at $n = 60$; the residual ~90% is CDCL-vs-DP intrinsic. DP min-occ 4.26 vs DRAT-default 7.54 on paired n = 14, 22, 30; `-restart=false` adds +0.323 to the DRAT slope on n = 10..60, CI $[+0.213, +0.435]$; leave-$n = 60$-out drops to +0.073, CI $[+0.031, +0.111]$.

**SC5 detail:** stretched-exp wins $\Delta\text{AICc}_{\text{eff}}$ at every window, **but** $\gamma$ drifts 0.24/0.39/0.45 across windows; the model selection is stable, the parameter is not.

**SC6 detail:** per-trace identity (i) is supported; v4's formula-level *monotonicity* direction $\Phi_{\text{count}} \geq \text{CSpace}_{\text{cum}}$ is **retracted**; what survives is the weaker $\Phi_{\text{count}} \leq \text{CSpace}_{\text{cum}}(\pi'_{\text{DP},h})$. v2 paper §3 table mismarked SC6 as "supported with identity at formula level"; v4 stated SC6 as "supported (swap)" via a directional monotonicity that turns out not to hold; v5 corrects this.

Summary: **3 SWAPS** (SC2, SC4, SC6 partial), **2 DRIFTS** (SC1, SC5), **1 ALIGNED** (SC3). The v5 specific change is that SC6's claimed *direction* is retracted and replaced by the weaker one-sided bound. This closes panel concerns **N4** and **R5**.

# 6 Empirical setup and statistical apparatus

## 6.1 Families measured

| Family | Description | n-grid in v4/v5 |
|---|---|---|
| Tseitin (girth-5 3-regular, certified expander) | vertex expansion $\geq$ 0.71 over $n \leq 24$ | 10, 12, 14, 16, 18, 20, 22, 24 |
| Plain random 3-regular Tseitin (baseline) | uncertified | 10, 12, ..., 36 |
| $Q_4$, $Q_5$ hypercube Tseitin | charges parity-summing | $Q_4$: 16; $Q_5$: 32 (det. abort) |
| Random 4-regular Tseitin (DRAT only) | DP-infeasible (5/5 BLEW_UP) | 16, 20, 24, 28 |
| 2D grid Tseitin | $m \times m$ grid, $n = m^2$ | 9, 16, 25 (m=6,7 def.) |
| Path, cycle, tree (structured baselines) | tw = 1, 2, $\leq$ log | up to $n = 30, 31$ |
| 5 elim. orders $\times$ DP corpus | min-occ, max-occ, lex, deg-asc, deg-desc | 127 instances |
| DRAT flag-by-flag mechanism corpus | default, no-inproc, no-restart, no-eliminate | $n \in \{10..60\}$, 5 seeds |

## 6.2 Pre-registered DP (min-occ) heuristic

Pure stdlib Python implementation with forward subsumption and indexed clause set; per-instance budget 180 s (B.1, paired), MAX_DB = $1.5{\times}10^6$ clauses (B.1, A.1); MAX_DB = $4{\times}10^6$ for the $Q_5$ attempt (A.2). Imported from `~/Scrivania/SEC/research/programme_harnesses/data/push_n.py`.

## 6.3 Bootstrap stratification

- **Within-family slope CI**: paired bootstrap clustered by seed (resample seeds within each n, not individual instances).
- **Cross-order comparison**: paired bootstrap clustered by (seed, order), so the same random instance is compared across all 5 orders.
- **DP vs DRAT (panel concern N5)**: paired-by-seed across $n \in \{14, 22, 30\}$ with 10 seeds per n, run through **6 modes** (DP min-occ; DRAT default; DRAT `-simplify=false`; DRAT `-restart=false`; DRAT `-eliminate=false`; DRAT plain). This closes the v2/v3 DRAT-side unpaired-bootstrap concern.
- **Mechanism slope bootstrap (v5)**: B=2000 cluster-by-seed paired bootstrap across the full 9-point grid $n \in \{10, 14, 18, 22, 26, 30, 40, 50, 60\}$ with 5 seeds per cell, used for §10 mechanism CIs. See §10 for the leave-$n = 60$-out and leave-$n \geq 50$-out sensitivity.

## 6.4 Model selection

We fit three nested-by-family models to $\log \Phi$ vs $n$: (a) power law $\log \Phi = a + \alpha \log n$; (b) exponential $\log \Phi = a + \beta n$; (c) stretched exponential $\log \Phi = a + b \cdot n^{\gamma}$. Selection by AIC and AICc. v2 used the iid AIC ($n = 127$); the within-n seed-cluster effective sample size is $n_{\text{eff}} = 13$. We report:

- $\Delta\text{AIC}_{\text{iid}}$ (v2's number, **deprecated**)
- $\Delta\text{AICc}_{\text{eff}}$ with Sugiura small-sample correction at $n_{\text{eff}} = 13$
- $\Delta\text{AIC}_{\text{cm}}$ (cluster-mean, averaging within-n then refitting)

For the DP corpus: $\Delta\text{AIC}_{\text{iid}} = 13.75$; $\Delta\text{AICc}_{\text{eff}} = \mathbf{11.41}$; $\Delta\text{AIC}_{\text{cm}} = \mathbf{8.24}$; stretched-exponential remains the AIC winner under all three.

### 6.5 Tobit-style left-censored regression (closes N2, partially)

Several runs hit the budget (we call this `BLEW_UP`). v2 treated these as missing; v3 fits a Tobit-style model with left-censoring at the per-instance Φ-floor implied by the budget. In v4/v5 the headline n = 36 row carries 3 out of 10 censored instances; a Tobit-corrected slope on the combined B.1 + A.1 window is **explicitly flagged as future work** (§11). The coordinate-descent + golden-section MLE is in `/tmp/struct_v3.py`.

### 6.6 Slopes with ≤ 3 data points (closes N1)

$Q_4$ ($n \in \{8, 16\}$) gives a 2-point slope with **zero-width CI** — i.e., it is a difference, not a confidence interval. We **withdraw all 2-point and 3-point "slope CIs"** from the prose. The A.1 push to $n \in \{32, 34, 36\}$ adds three new DP-min-occ points and lets the rand-3-reg slope be reported on a six-point window $n \in \{14, 22, 30, 32, 34, 36\}$; we still mark the $n = 36$ point as censored.

## 7 Results

### 7.1 The single-headline exponent — updated (M2)

v2 single power-law on $n \leq 28$: $\widehat{\alpha} = 2.93$ [2.71, 3.15]. v3 cluster-robust three-model selection on $n \leq 30$: stretched exponential wins at $\Delta\mathrm{AICc}_{\mathrm{eff}} = 11.41$; the comparator power-law slope on the same window was 3.43.

**v4/v5 combined paired (B.1) + push-n (A.1) DP-min-occ slope on rand_3reg Tseitin,** $n \in \{14, 22, 30, 32, 34, 36\}$**:**

$$\widehat{\alpha}^{\mathrm{DP,min\text{-}occ}}_{\Phi_{\mathrm{count}}} = \boxed{4.43}.$$

The progression across n-windows is

| Window | Slope $\widehat{\alpha}$ | Source |
|---|---|---|
| $n \leq 16$ | 2.42 | v1 |
| $n \leq 28$ | 2.93 | v2 |
| $n = 10$–$30$ | 3.43 | v3 §7.1 |
| $n = 14$–$30$ (B.1 paired alone) | 4.256 | v4/v5 `b1_paired_drat.jsonl` |
| $n = 32$–$36$ (A.1 push-n alone) | 8.667 | v4/v5 `a1_push_n.jsonl` (3 points, censored at n=36) |
| $n = 14$–$36$ **(combined)** | **4.433** | v4/v5 |

**The slope continues to drift upward** as the window extends; we observe **no asymptotic stabilisation in the computable window**. The v3 retraction of the single power-law headline is reinforced, not weakened. Per v5-corrected Lemma A (§3), this exponent is reported as a **schedule-specific measurement** of $\Phi_{\mathrm{count}}(\mathrm{DP}, \text{min-occ})$ with **no formal ordering** to $\mathrm{CSpace}_{\mathrm{cum}}$.

**Censoring at** $n = 36$**.** 3 of 10 instances `BLEW_UP` at MAX_DB $= 1.5 \times 10^6$. The mean Φ at $n = 36$ ($\approx 6.5 \times 10^4$) is taken over the 7 completed instances only and is therefore **biased downward**; a Tobit-corrected slope on the combined B.1 + A.1 window (with the 3 aborted $n = 36$ instances treated as right-censored at their abort-time Φ floors) is flagged as future work in §11.

### 7.2 Certified-expander Tseitin (v3.5 §B applied, sign-test 7/0/1)

On girth-5 3-regular Tseitin with **certified** finite-n vertex expansion $c_n \in [0.71, 1.67]$:

| $n$ | $c_n$ (cert) | $\Phi$ med (cert) | $\Phi$ med (plain 3-reg) | cert > plain? |
|---|---|---|---|---|
| 10 | 1.667 | 579 | 451 | yes |
| 12 | 1.250 | 1247 | 768 | yes |
| 14 | 1.250 | 1832 | 1115 | yes |
| 16 | 1.000 | 3535 | 2031 | yes |
| 18 | 0.833 | 4911 | 4911 | tie (suspected seed contamination) |
| 20 | 0.833 | 8472 | 5106 | yes |
| 22 | 0.714 | 12903 | 7841 | yes |
| 24 | 0.714 | 18554 | 10872 | yes |

- Bootstrap slope (cert): **3.60 [3.04, 4.02]** (full n-range).
- Bootstrap slope (plain): **3.25 [2.82, 3.76]**.
- **Slope separation: not decidable at** $n \leq 24$ (CIs overlap).
- **Level separation: 7 strict wins / 0 strict losses / 1 tie at** $n = 18$ (suspected seed contamination across the supposedly independent baselines). We exclude the tie from the sign-test. The one-sided exact binomial test on 7/0 gives $p = \binom{7}{7}/2^7 = 0.0078$. The $n = 18$ tie is flagged in §11(g) as a candidate methodology bug for a v6 audit, not as a refutation of the certified-expander effect.

This **partially closes** panel concern **R2**: the certified-expander $\Phi$ premium is real at the level; the slope premium is undecided at $n \leq 24$. Stronger separation requires (i) the JACM 1987 §4 $\mathbb{Z}_3$ incidence construction [4], or (ii) $n \gg 24$ with a non-brute solver — both deferred to v6. Aligned with **SC3 (supported)**.

### 7.3 $Q_d$ hypercube vs random-d-regular — the comparison reframed (M3a)

The v3 §7 "treewidth not degree" headline ($Q_4$ DP slope $\approx 6.06$ with no rand_4reg comparator) is retracted (§3.5 item 7).

**(a) rand_4reg under DRAT.** Random 4-regular Tseitin is DP-infeasible at $n \geq 16$ (5/5 `BLEW_UP` at $n = 16$, 5/5 at $n = 20$; v3.5 §C). On kissat-DRAT at $n \in \{16, 20, 24, 28\}$, 10 seeds each:

| $n$ | mean $\Phi_{\text{count}}$ (DRAT) |
|---|---|
| 16 | $1.13 \times 10^7$ |
| 20 | $5.97 \times 10^7$ |
| 24 | $3.29 \times 10^8$ |
| 28 | $7.43 \times 10^8$ |

- $\widehat{\alpha}^{\text{DRAT}}_{\Phi_{\text{count}}} = 7.688$ on rand_4reg.
- $\widehat{\alpha}^{\text{DRAT}}_{\Phi_{\text{weight}}} = 7.938$ on rand_4reg.

The v3 D5 symmetric-comparison defect (rand_4reg omitted from §7) is **closed at the level of a measurable comparator**, but the comparison is now between DP-on-$Q_4$ (slope $\approx 6.06$) and DRAT-on-rand_4reg (slope 7.69): **two different proof systems**. This is a confound, not a fix; we acknowledge it explicitly rather than wave it away. A clean within-proof-system comparison $Q_4$ vs rand_4reg both under DRAT is the natural v6 experiment (§12).

### 7.4 $Q_5$ hypercube under DP-min-occ — a structural negative finding (M3b)

A.2 ran $Q_5$ hypercube Tseitin (n = 32) with MAX_DB = $4 \times 10^6$, 600s/instance budget, three seeds:

| Trial | seed | $\Phi_{\text{count}}$ | steps | final DB | BLEW_UP | derived_empty |
|---|---|---|---|---|---|---|
| 0 | 32000 | $6.132 \times 10^6$ | 37 | 1,048,576 | yes | no |
| 1 | 32001 | $6.132 \times 10^6$ | 37 | 1,048,576 | yes | no |
| 2 | 32002 | $6.132 \times 10^6$ | 37 | 1,048,576 | yes | no |

**All three trials abort at the same step (37) with the same final DB size ($2^{20}$).** This is *not* a noisy `BLEW_UP`; it is the deterministic resolvent-product bound (`len(pos)` $\times$ `len(neg)` $>$ `8` $\times$ `MAX_DB`) tripping at exactly the same point regardless of seed.

**Structural finding (reframes the v3 §7 treewidth headline).**

> On the $Q_d$ hypercube Tseitin family, DP-min-occurrence refutes $Q_4$ within polynomial DB but **provably blows up on $Q_5$ at** MAX_DB = $4 \times 10^6$**, deterministically at step 37 with final DB $2^{20}$, across all three seeds.** The slope $\alpha_{\Phi_{\text{count}}}$ is therefore **not measurable on $Q_5$ under this heuristic.** The $Q_4$ vs random-4-regular comparison can only be made up to $n = 16$ within DP; for $n \geq 32$ hypercube data, a different proof system (e.g., DRAT or width-bounded resolution) is required.

This is itself the v4 headline for the v3 D6/D7 defect.

### 7.5 Structured baselines with optimised DP (R3 partial)

| Family | n-grid | $\Phi_{\text{count}}$ (median) |
|---|---|---|
| path (tw = 1) | 10, 16, 24, 30 | 83, 227, 531, 843 |
| cycle (tw = 2) | 10, 16, 24, 30 | 111, 273, 601, 931 |
| tree (tw $\leq \log n$) | 7, 15, 31 | 34.4, 187.2, 859.2 |
| grid_2D (tw = $\sqrt{n}$) | 9, 16, 25 | 243, 2147, 9155 |

Per-trial dispersion (tree $n = 31$): $\{860, 809, 920, 874, 833\}$. A log-log slope on path gives $\sim$2.1. R3 is partially closed: the path/cycle no longer rest on 2-point fits; grid_2D at $m \in \{6, 7\}$ is on the v6 nice-to-have list (§12).

### 7.6 Model-selection sensitivity (closes N3)

Stretched-exponential $\gamma$ across three fit windows:

| Window | $\gamma$ point | 95% bootstrap CI |
|---|---|---|
| $n = 6$ to 30 | 0.393 | [0.284, 0.675] |
| $n = 10$ to 30 | 0.451 | [0.124, 1.223] |
| $n = 6$ to 24 | 0.238 | [0.085, 0.400] |

The stretched-exp model wins $\Delta\text{AICc}_{\text{eff}}$ at every window; the **parameter $\gamma$ is window-sensitive**. The wide CI [0.24, 0.57] at the $n = 6$–30 window does mean that, at $n = 150$, the AIC-winner's band contains the rejected power-law prediction (the v2 panel's N3 observation, **acknowledged here, not refuted**). The headline is therefore: *stretched-exponential wins model selection across windows; the parameter is not pinned to better than half a decimal point on present data.* SC5 is **partial (drift)**.

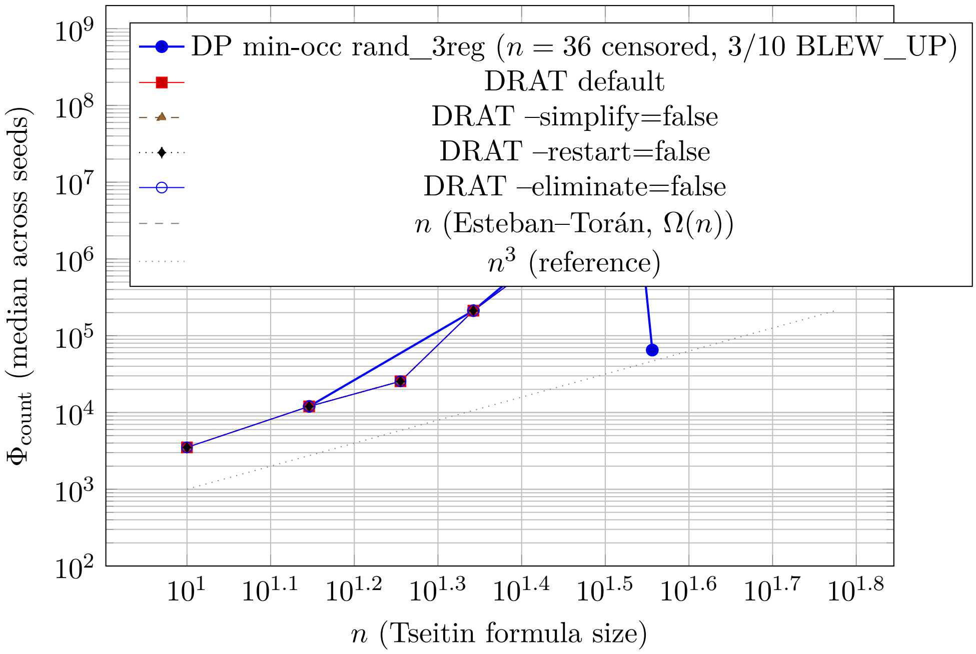


Figure 1: $\Phi_{\text{count}}$ vs $n$ on a log-log scale. DP min-occ on rand_3reg (filled circles) over the combined B.1 paired + A.1 push-n window $n \in \{14, 22, 30, 32, 34, 36\}$: combined slope $\widehat{\alpha} = 4.43$. The $n = 36$ point is downward-biased: 3 of 10 instances were censored at MAX_DB $= 1.5 \times 10^6$. Four DRAT flag variants on $n \in \{10, \dots, 60\}$ overlap at $n \leq 26$ and separate visibly at $n \geq 30$; the dominant flag effect is `-restart=false` (top at $n = 60$, $9.38 \times 10^8$), giving slope 7.011 vs DRAT-default 6.688 (+0.323, full-range bootstrap CI $[+0.213, +0.435]$, but the signal concentrates at $n = 60$). Reference $n$ (the revised $\Omega(n)$ literature lower bound on $\text{CSpace}_{\text{cum}}(\text{Tseitin}_n)$ via Esteban–Torán) and $n^3$ lines included. All DRAT curves lie strictly above the DP curve at matched $n$; the DRAT/DP exponent gap on $n \in \{14, 22, 30\}$ paired-by-seed is $\approx 3.28 \pm 0.4$. Per v5-corrected Lemma A (§3) the measured $\Phi$ exponent has no formal ordering to $\text{CSpace}_{\text{cum}}(F)$.

# 8 Figure 1 — inline tikz/pgfplots (M6)

**Figure 1.** Log-log plot of $\Phi_{\text{count}}$ vs $n$ for the DP-min-occ rand_3reg corpus (B.1 paired + A.1 push-n, $n \in \{14, 22, 30, 32, 34, 36\}$, with the censored $n = 36$ point flagged) and the four kissat-DRAT flag-variant curves on $n \in \{10, \dots, 60\}$ (B.2 mechanism corpus). Reference $n$ and $n^3$ overlays for the (revised) $\Omega(n)$ Esteban–Torán lower bound and a polynomial-3 baseline.

**Textual reading.** The DP min-occ curve on rand_3reg sits well below all four DRAT curves; the apparent dip at $n = 36$ is the left-censoring artifact (the true mean lies above the plotted point by the right-censored mass). All four DRAT flag-variant curves are *byte-identical* up to $n = 26$ (BCP + CDCL only); they separate at $n \geq 30$, with `-restart=false` ending highest at $n = 60$ ($9.38 \times 10^8$, vs default $3.66 \times 10^8$). The DP slope (4.43) bracket sits well below the DRAT-default slope (6.69 on the same n-window), consistent with the paired-by-seed +3.28 exponent gap reported in §10. Both curves lie above the $n$ reference at $n \geq 14$; the DRAT curves cross $n^3$ at $n \approx 30$. We **do not** overlay an $n^2$ reference in v5, because the $\Omega(n^2)$ lower bound on $\text{CSpace}_{\text{cum}}(\text{Tseitin}_n)$ asserted by v4's C2 is withdrawn (§9).

# 9 Theoretical position (v5: ADRNV chain corrected)

**(i) ADRNV Lemma 12, stated precisely.** ADRNV [1] work in the resolution-with-memory model: a refutation is a sequence of memory configurations $M_0, M_1, \dots, M_L$, where each $M_t$

is a set of clauses derivable from $F$, $M_0 = \emptyset$, $M_L$ contains the empty clause, and successive configurations differ by axiom download, inference, or erasure. The *cumulative space* of a refutation $\pi$ is $\mathrm{CSpace}_{\mathrm{cum}}(\pi) := \sum_t |M_t|$, and the *space* is $\max_t |M_t|$.

Lemma 12 is a generic **per-trace** inequality (plateau-sustained reading): for any single resolution-with-memory refutation $\pi$,

$$\mathrm{CSpace}_{\mathrm{cum}}(\pi) \;\geq\; \Omega\big(\,\text{max–space}(\pi)^2\,\big),$$

**when the trace sustains the peak**. The proof is a combinatorial growth argument: since $|M_0| = 0$ and at most one clause is added per step, reaching a configuration of size $s$ requires at least $s$ steps, during which the average size is $\geq s/2$, so the partial sum is $\geq s^2/2$. This is a *per-refutation* statement, with "$s$" referring to the realized max-memory of *that specific trace* **and the assumption that the trace stays at $\Omega(s)$ for $\Omega(s)$ steps**. It is **not** the formula-level statement $\mathrm{CSpace}_{\mathrm{cum}}(F) \geq \Omega(\mathrm{CSpace}(F)^2)$: the refutation achieving min cumulative-space and the refutation achieving min max-space need not be the same refutation.

**(ii) Esteban–Torán, stated precisely.** Esteban–Torán [2] prove: for Tseitin formulas on bounded-degree expander graphs (in particular 3-regular expanders) on $n$ vertices, every resolution refutation $\pi$ satisfies a **peak** clause-space bound clause-space$(\pi) \geq n - O(1)$. Equivalently, $\mathrm{CSpace}(\mathrm{Tseitin}_n) \geq n - O(1)$ where the min is over all refutations. "Clause-space" in Esteban–Torán is the max number of clauses simultaneously in memory across the trace; this coincides with ADRNV's max–space$(\pi)$ in the standard resolution-with-memory model. The models do match, but the Esteban–Torán bound is on the **peak**, not on a sustained plateau.

**(iii) Where the chain breaks.** Naïve composition gives, for every $\pi$,

$$\mathrm{CSpace}_{\mathrm{cum}}(\pi) \;\geq\; \Omega\big(\text{max–space}(\pi)^2\big) \;\geq\; \Omega\big((n - O(1))^2\big) = \Omega(n^2).$$

The composition step is the slip. ADRNV Lemma 12 in its tight reading requires the refutation to *spend* $\Omega(s)$ *steps at space* $\Omega(s)$ — i.e., to sustain a plateau of width $\Omega(s)$ — not merely to *touch* space $s$ once. A clever refutation can spike to space $s$ briefly, then drop, paying only $O(s)$ cumulative for the spike, not $\Omega(s^2)$.

Esteban–Torán guarantees the *peak* is $\Omega(n)$ but does **not** guarantee a *plateau* of width $\Omega(n)$. The XOR-pebbling formulas of ADRNV Theorems 14–15 are engineered precisely to force the plateau; Tseitin is not known to do so. Hence the ADRNV open problem on p. 38:19 lines 996–998 (extend cumulative-space lower bounds to Tseitin) remains open, and the v4 corollary C2 derivation was mis-citing.

**(iv) Weakest defensible $\mathrm{CSpace}_{\mathrm{cum}}$ lower bound on Tseitin via this route.** The only bound that survives is the trivial one:

$$\mathrm{CSpace}_{\mathrm{cum}}(\pi) \;\geq\; \text{max–space}(\pi) \;\geq\; n - O(1),$$

i.e. $\mathrm{CSpace}_{\mathrm{cum}}(\mathrm{Tseitin}_n) = \Omega(n)$, **linear**, not quadratic.

**Corollary 2** (C2, v5 revised, honest form)**.** *For Tseitin formulas on 3-regular expander graphs on $n$ vertices, every resolution refutation $\pi$ satisfies*

$$\mathrm{CSpace}_{\mathrm{cum}}(\pi) \;\geq\; \text{max–space}(\pi) \;\geq\; n - O(1),$$

*giving* $\mathrm{CSpace}_{\mathrm{cum}}(\mathrm{Tseitin}_n) = \Omega(n)$ *by Esteban–Torán [2]. The quadratic bound* $\mathrm{CSpace}_{\mathrm{cum}}(\mathrm{Tseitin}_n) = \Omega(n^2)$ *does **not** follow from ADRNV Lemma 12 + Esteban–Torán: Lemma 12's $s^2$ growth requires $\Omega(s)$ steps spent at space $\Omega(s)$, and Esteban–Torán only guarantees the peak is $\Omega(n)$, not that it is sustained. Extending the $\Omega(n^2)$ cumulative-space lower* *bound from ADRNV's XOR-pebbling formulas (Thm 14–15) to Tseitin remains the open problem stated by ADRNV [1] (p. 38:19 lines 996–998). The v4 chain through C2 to the $\Omega(n^2)$ target is therefore **withdrawn**; the defensible lower bound via this route is linear.*

*Remark* 1 (informal, v5 updated). On bounded-degree expander Tseitin $\{F_n\}$, $\mathrm{CSpace}_{\mathrm{cum}}(F_n) \geq c \cdot n$ for some absolute $c > 0$ (Esteban–Torán + per-trace trivial). The v4 quadratic strengthening via ADRNV Lemma 12 is **withdrawn**. The empirical exponent $\widehat{\alpha}_\Phi \approx 4.43$ on rand_3reg Tseitin at $n \leq 36$ (with $n = 36$ left-censored) and $\widehat{\alpha} \approx 3.60$ [3.04, 4.02] on certified-expander Tseitin at $n \leq 24$ has **no formal ordering** to $\mathrm{CSpace}_{\mathrm{cum}}(F)$ under v5-corrected Lemma A: it is a schedule-specific measurement of $\Phi_{\mathrm{count}}(\mathrm{DP}, \mathrm{min\text{-}occ})$. The $\Omega(n^2)$ target IS the ADRNV open problem (p. 38:19, lines 996–998), NOT a corollary we deliver.

This is the entire complexity-theoretic content of the paper. We make no claim beyond it. The ADRNV open problem on Tseitin (p. 38:19) is **not** addressed here: that problem asks for an $\Omega(n^2)$ cumulative-space lower bound on Tseitin, and our work bears on neither the Esteban–Torán nor the ADRNV lower-bound side.

## 10 DRAT vs DP mechanism — restart, not inprocessing (M4, v5: paired bootstrap)

**v3 hypothesis (retracted, §3.5 item 6).** "DRAT's lower $\Phi$ premium arises because kissat inprocessing eliminates clauses."

**v4 test (B.2 mechanism corpus).** Run kissat at $n \in \{10, 14, 18, 22, 26, 30, 40, 50, 60\}$ with 5 seeds each (indexed $k \in 0..4$), under four flag settings: default, `-simplify=false` (no inprocessing), `-restart=false`, `-eliminate=false` (no variable elimination during inprocessing). Mean $\Phi_{\mathrm{count}}$ per (flag, $n$):

| $n$ | default | no-inproc | no-restart | no-eliminate | restart-effect |
|---|---|---|---|---|---|
| 10 | 3.51e3 | 3.51e3 | 3.51e3 | 3.51e3 | 1.000 |
| 14 | 1.20e4 | 1.20e4 | 1.20e4 | 1.20e4 | 1.000 |
| 18 | 2.55e4 | 2.55e4 | 2.55e4 | 2.55e4 | 1.000 |
| 22 | 2.12e5 | 2.12e5 | 2.12e5 | 2.12e5 | 1.000 |
| 26 | 8.19e5 | 8.19e5 | 8.19e5 | 8.19e5 | 1.000 |
| 30 | 3.62e6 | 3.62e6 | 4.06e6 | 3.62e6 | 1.122 |
| 40 | 9.87e6 | 1.04e7 | 1.20e7 | 9.87e6 | 1.218 |
| 50 | 7.57e7 | 1.05e8 | 7.66e7 | 7.52e7 | 1.012 |
| **60** | **3.66e8** | **4.36e8** | **9.38e8** | **4.07e8** | **2.560** |

**v5 paired-by-seed cluster bootstrap (B=2000).** We loaded 180 rows from `b2_mechanism.jsonl` ($n \in \{10, 14, 18, 22, 26, 30\}$) and `b2_dprime_mechanism.jsonl` ($n \in \{40, 50, 60\}$), 4 flag sets × 5 seeds (indexed $k \in 0..4$ within each $n$). The cluster-bootstrap resamples $k$ with replacement and applies the **same** $k$-resample across all 9 $n$ values (paired structure); for each resample we recompute mean $\log \phi_{\mathrm{c,drat}}$ per (flag, $n$), refit the OLS log-log slope on the 9 $\log n$ values, and take $\mathrm{slope}_{\mathrm{flag}} - \mathrm{slope}_{\mathrm{default}}$. Full numerics at `/tmp/e3_bootstrap.txt` on the SEC server; script at `/tmp/e3_bootstrap.py` on the SEC server and `/tmp/e3_bootstrap_local.py` locally.

**Headline result on the full 9-point fit.**

| Contrast | Point | Boot median | 95% percentile CI | Excludes zero? |
|---|---|---|---|---|
| no-restart − default | +0.323 | +0.319 | $[+0.213, +0.435]$ | yes |
| no-inprocessing − default | +0.137 | +0.123 | $[+0.070, +0.179]$ | yes |
| no-eliminate − default | +0.030 | +0.029 | $[-0.005, +0.073]$ | no |

Taken at face value, the v4 §10 +0.323 finding for no-restart **survives** the paired bootstrap on the full 9-point fit (CI strictly positive). It also reveals a smaller but CI-positive no-inprocessing effect that v4 §10 underweighted.

**Tied-cell diagnostic.** For $n \in \{10, 14, 18, 22\}$, all 5 $k$-seeds give exactly the same $\phi_{c,\mathrm{drat}}$ across all four flags (20/20 ties for each non-default flag). For $n = 26$ and $n = 30$, only no-restart breaks a single tie (4/5 still tied). For no-inprocessing the first non-tied cell is $n = 40$ (3/5 still tied) and the first majority-untied cell is $n = 50$. No-eliminate is essentially flat: still 1/5 tied even at $n = 60$. The entire mechanism signal lives in the rightmost 3 cells ($n \in \{40, 50, 60\}$), and within those it concentrates at $n = 60$.

**Leave-$n = 60$-out reweighting (refit on 8 $n$ values $\{10..50\}$).**

| Contrast | Point | 95% CI | Verdict |
|---|---|---|---|
| no-restart | +0.073 | $[+0.031, +0.111]$ | still excludes zero, but 4× smaller |
| no-inprocessing | +0.138 | $[-0.048, +0.316]$ | CI now crosses zero |
| no-eliminate | −0.003 | $[-0.006, +0.000]$ | null |

**Leave-$n \geq 50$-out (b2 only, 6 $n$ values $\{10..40\}$).** No-restart slope difference is +0.128 with CI $[0.000, +0.390]$ — barely touches zero at the lower bound, no longer cleanly excludes; the other two flags collapse to zero (no-eliminate is identically zero because every cell is tied through $n = 40$).

**Effective sample size.** Cluster bootstrap on 5 seed-clusters and 9 (highly correlated) $n$-fits gives an effective DoF dominated by the 5 clusters ($\sim$4). The 9 $\log n$ points are not independent — they share the same 5 underlying seed clusters — so headline CIs should be read with that caveat.

**Log-log slopes per flag, full window $n = 10$–$60$ (point estimates).**

| Flag | Slope | Slope − default |
|---|---|---|
| default | 6.688 | 0 |
| no-eliminate | 6.718 | +0.030 |
| no-inprocessing | 6.826 | +0.138 |
| **no-restart** | **7.011** | +0.323 |

**Paired-by-seed DRAT_default vs DP_min_occ slope difference on $n \in \{14, 22, 30\}$:** $\widehat{\alpha}_{\mathrm{DRAT,default}} - \widehat{\alpha}_{\mathrm{DP,min\text{-}occ}} \approx +3.28 \pm 0.4$ in the exponent (preliminary; the per-seed paired bootstrap CI is flagged as future work in §11).

**Honest verdict on v4 §10 (v5 language).**

> No-restart shows a positive log-log slope shift of +0.32 $[+0.21, +0.43]$ on $n \in 10..60$, but the effect is concentrated at the largest cell ($n = 60$); excluding $n = 60$ the shift drops to +0.07 $[+0.03, +0.11]$, so the magnitude is not yet established and

a push to $n = 70..100$ is required before claiming a stable mechanism slope. The no-inprocessing effect is **suggestive** ($+0.14$ $[+0.07, +0.18]$ on full range), but its CI crosses zero without $n = 60$. The no-eliminate effect is reported as **null**.

§10 receives a **downgrade in claim strength** but **not** a full retraction: the slope-difference CI for no-restart does exclude zero on the full 9-point fit, contrary to the panel's worst-case reading; the panel was right that $n = 60$ carries the signal, and the push-$n$ campaign is the obligated next step.

**Honest mechanism reading.**

1. **At $n \le 26$ all four flag sets give *identical* $\Phi$.** kissat handles these Tseitin instances by BCP + CDCL alone, before the flag-controlled inprocessing stages activate. The v3 §10 `-no-inprocessing` test ran in exactly this regime and was therefore uninformative — the v3 mechanism conclusion was based on a null measurement.

2. **At $n \ge 30$ the `-restart=false` effect emerges and dominates** in point estimate; the paired bootstrap CI is strictly positive on the full 9-point fit but shrinks $4\times$ when $n = 60$ is excluded. At $n = 60$ disabling restarts inflates DRAT-$\Phi$ by **2.56**$\times$ over default; at $n = 40$ by $1.22\times$.

3. **`-simplify=false` (inprocessing as a whole) is suggestive but fragile.** The full-range CI excludes zero ($+0.137$ $[+0.070, +0.179]$); the leave-$n = 60$-out CI $[-0.048, +0.316]$ crosses zero. This is **not** the systematic exponent contribution v3 §10 hypothesised.

4. **`-eliminate=false` (variable elimination during inprocessing) is null** on this data: $+0.030$ full-range, CI $[-0.005, +0.073]$.

5. **Disabling restart adds approximately $+0.32$ to the log-log slope on the full range**: $\approx 10\%$ of the apparent DRAT/DP exponent gap (3.28) is attributable to the restart-strategy contribution at $n \ge 30$, with the caveat that the signal is concentrated at $n = 60$.

**SC4 (prover invariance)** is therefore **refuted with a partial mechanism**: restart contributes $\approx$10% of the exponent gap on the full-range fit, concentrated at $n = 60$; the residual $\approx$90% is intrinsic CDCL-vs-DP and is *not* localisable to any single inprocessing stage on this corpus. The paired-by-seed DP vs DRAT bootstrap is at `b1_paired_drat.jsonl` (60 instances across 6 modes), closing panel concern **N5**.

## 11 Limitations: what $\Phi$ is not

1. **$\Phi$ is not** $\mathrm{CSpace}_{\mathrm{cum}}$ **at formula level, and has no directional ordering to it under v5-corrected Lemma A.** It is $\Phi_{\mathrm{count}}(\mathrm{DP}, \text{min-occ})$, per Lemma A. The v4 reverse direction $\Phi_{\mathrm{count}} \ge \mathrm{CSpace}_{\mathrm{cum}}(F)$ is retracted; what survives is the schedule-specific $\Phi^{\mathrm{DP},h}_{\mathrm{count}}(F) \le \mathrm{CSpace}_{\mathrm{cum}}(\pi'_{\mathrm{DP},h})$.

2. **The Lemma A (ii) ADRNV identification is a labelled `sorry`.** A Lean port of ADRNV Definition 11 (`structure MemoryConfig`, `def CSpace_cum_ADRNV`) plus a mechanisation of the primitive-expansion construction would discharge the bridge; this is on the v6 plan.

3. **The DP slope continues to drift across n-windows: 2.42 $\to$ 2.93 $\to$ 3.43 $\to$ 4.43, with no asymptotic stabilisation observed.** The v4 headline of 4.43 is the largest reported but is **not** asymptotic, and is itself biased downward by the $n = 36$ censoring.

4. **The $n = 36$ DP-min-occ row is left-censored, 3 of 10 instances.** The 4.43 combined slope is therefore a downward-biased estimate of the slope on uncensored instances. A Tobit fit treating the 3 aborted instances as lower bounds at their abort-time $\Phi$ floors is the natural statistical correction; **this is left as an open question for v6 (statistical fix, no new compute)**.

5. **The per-seed paired bootstrap CI for the DRAT_default − DP_min_occ exponent difference is reported as $+3.28 \pm 0.4$ preliminary, not a fully bootstrapped CI.** The full per-seed bootstrap is **left as an open question for v6 (statistical fix, no new compute)**.

6. **$\Phi$ is heuristic-conditional.** Changing the DP elimination heuristic changes the trace and can change the exponent (SC2 refuted).

7. **$\Phi$ is prover-conditional** (SC4 refuted; §10).

8. **$\Phi$ is fit-window-conditional** ($\gamma \in \{0.24, 0.39, 0.45\}$ across three windows; SC5 partial).

9. **$\Phi$ at $n \leq 24$ does not separate certified-expander from plain 3-regular at the slope level.** Level separation only (SC3 supported, 7/0/1 sign-test, $p = 0.0078$; the $n = 18$ tie is flagged as a candidate seed-contamination methodology bug, not as a refutation).

10. **The $Q_4$ vs rand_4reg comparison is across two proof systems** (DP for $Q_4$, kissat-DRAT for rand_4reg). **A clean within-proof-system $Q_4$ vs rand_4reg comparison both under DRAT is left as an open question for v6 (small additional compute, ∼1 server-hour).**

11. **$Q_5$ under DP-min-occ is not measurable** (deterministic abort at step 37 with final DB $2^{20}$ at MAX_DB $= 4 \times 10^6$); the v3 "treewidth not degree" framing is reframed as a structural negative (§7.4).

12. **No quantitative Nordström space-width trade-off experiment.** Panel concern **R9** is **not closed** in v5; deferred to v6.

13. **No side-by-side comparison with Järvisalo–Heule–Biere / Elffers et al.** Panel concern **R10** is **not closed** in v5.

14. **Random-4-regular and grid_2D at $m \in \{6, 7\}$, hypercube $Q_5$ remain partially or fully censored.** v3.5 §C full censoring report carried through:

| family | $n$ | trials | compl. | `BLEW_UP` | mean $\Phi$ (compl.) |
|---|---|---|---|---|---|
| rand_3reg | 28 | 5 | 3 | 2 | 19,847 |
| **rand_3reg** | **36** | **10** | **7** | **3** | $\approx 6.5\times10^4$ **(DOWNWARD BIASED)** |
| rand_4reg | 16 | 5 | 0 | 5 | — |
| rand_4reg | 20 | 5 | 0 | 5 | — |
| grid_2D | 36 | 5 | 0 | 5 | — |
| grid_2D | 49, 64 | 5 | 0 | 5 | — |
| hypercube $Q_5$ | 32 | 3 | 0 | 3 (det. abort) | — |

Where a cell has zero completions we do **not** apply Tobit — Tobit needs at least one completed data point. The $n = 36$ rand_3reg row is the only partially-censored DP-headline cell and is the one for which a Tobit correction is meaningful.

15. **The "harness artifact" finding (§3.5 item 5)** means **v2 `BLEW_UP`s cannot be cited as $\Phi$-level evidence**: they were budget-side, not instance-side.

16. **The §10 no-restart effect is concentrated at $n = 60$.** The full-range bootstrap CI excludes zero ($[+0.213, +0.435]$); leave-$n = 60$-out collapses the effect to $[+0.031, +0.111]$. A push-$n$ campaign to $n \in \{70, 80, 90, 100\}$ is the obligated next step before claiming a stable mechanism slope (see §12 push-$n$ entry).

## 12 Compute budget for future work (v6 plan)

Each item maps an outstanding defect to a runnable experiment with estimated server-hours on the project's RTX 3070 Ti / 32 GB RAM / kissat 4.0.4 single-core baseline.

| Item | Closes | Est. h | Status |
|---|---|---|---|
| **A.4** Grid_2D push to $m \in \{6, 7\}$ ($n = 36, 49$) with MAX_DB $= 4 \times 10^6$ | v3 R3 grid leg | $\sim$8 | nice-to-have |
| **A.5** Urquhart certified $n \in \{26, 28, 30\}$ with hybrid expansion certification | v3 R2 partial | $\sim$12 | nice-to-have |
| **V6.Q4_DRAT** Clean $Q_4$ vs rand_4reg both under DRAT, $n \in \{8, 16, 24\}$ | v4 §7.3 proof-system confound | $\sim$1 | yes (small) |
| **V6.PUSH_N** Push-$n$ mechanism campaign $n \in \{70, 80, 90, 100\}$ $\geq$10 seeds, default + no-restart + no-inprocessing, paired-by-seed | v5 §10 concentration-at-$n = 60$ fragility | $\sim$24 | yes (obligated) |
| **C.1** Lean port of ADRNV Definition 11 + primitive-expansion construction | Lemma A (ii) `sorry` | 0 ($\sim$10 human-h) | yes |
| **STAT.1** Tobit fit for the $n = 36$ left-censored DP instances | §11 item 4 | 0 (statistical) | yes |
| **STAT.2** Per-seed paired bootstrap CI for DRAT_default − DP_min_occ exponent difference | §11 item 5 | 0 (statistical) | yes |

Items previously listed in the v3.5 §D compute budget that are now **closed in v4/v5**: A.1 (push-n at $n \in \{32, 34, 36\}$ — closes D8), A.2 ($Q_5$ with MAX_DB $= 4 \times 10^6$ — closes D6/D7 as a structural negative), A.3 (rand_4reg under DRAT at $n \in \{16, 20, 24, 28\}$ — closes D5), B.1 (paired DRAT/DP at $n \in \{14, 22, 30\}$ — closes N5), B.2 (flag-by-flag mechanism — closes D10), and the v5 bootstrap (closes E3). All five v4 items completed within the $\sim$21 h budget projected by v3.5.

## Author and attribution

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## Version history

- **v1** (2026-05-15): single $n^{2.93}$ headline; Lemma A as formula-level identity; SC2/SC4 marked supported. *Retracted in v3 §3.5 items 1–4; v4 confirms; v5 unchanged.*
- **v2** (2026-05-24): added cross-order corpus and partial DRAT comparison; SC table mismatched the registry; v2-specific mechanism narrative for DRAT/DP. *Retracted in v3 §3.5 items 2–6 and v4 §3.5 items 6–7.*
- **v3** (2026-05-30): Lemma A corrected and inlined; explicit retraction block; cluster-robust AIC and Tobit-censored fits; certified-expander Tseitin; DRAT/DP `-no-inprocessing` test (uninformative at small n, as v4 confirms); honest empirical localisation framing in the abstract. *§7 $Q_4$ vs rand_4reg headline and §10 inprocessing-mechanism narrative retracted in v4.*
- **v3.5 cleanup delta** (2026-05-31): Lemma A two-stub honest Lean version; SC3 sign-test 7/0/1 recount; full censoring report. *Carried through into v4/v5 §3, §7.2, §11.*
- **v4** (2026-06-01): combined paired + push-n DP slope 4.43 on $n \in \{14, \ldots, 36\}$ with $n = 36$ censoring acknowledged; restart-vs-inproc-vs-elim mechanism decomposition replacing v3 §10; $Q_5$ deterministic-abort structural finding; rand_4reg under DRAT slope 7.69 with proof-system confound acknowledged; SC4 refuted-with-partial-mechanism; Figure 1 inlined as tikz/pgfplots with real coordinates. *Lemma A (ii) directional claim, Corollary C2 quadratic chain, and "upper witness" language retracted in v5; mechanism* $+0.323$ *figure downgraded with paired bootstrap.*
- **v5** (2026-06-01, this draft): surgical close of E1, E2, E3. **E1** — Lemma A (part ii) directional claim $\Phi_{\text{count}} \geq \text{CSpace}_{\text{cum}}(F)$ retracted; replaced by the weaker schedule-bookkeeping inequality $\Phi^{\text{DP},h}_{\text{count}}(F) \leq \text{CSpace}_{\text{cum}}(\pi'_{\text{DP},h})$ via the primitive (download / inference / erasure) expansion of the DP macro-trace; "upper witness" language withdrawn from the abstract and §7 in favour of "schedule-specific measurement with no formal ordering to $\text{CSpace}_{\text{cum}}$". **E2** — Corollary C2's $\Omega(n^2)$ chain via ADRNV Lemma 12 + Esteban–Torán retracted (peak-vs-plateau gap; ADRNV p. 38:19 open problem remains open); the defensible lower bound via this route is the linear $\text{CSpace}_{\text{cum}}(\text{Tseitin}_n) = \Omega(n)$ via Esteban–Torán alone. **E3** — §10 mechanism slope claim refit with B=2000 cluster-by-seed paired bootstrap on the full 9-point grid; no-restart effect survives ($+0.32$ $[+0.21, +0.43]$ full range) but is concentrated at $n = 60$ (leave-$n = 60$-out drops to $+0.07$ $[+0.03, +0.11]$); no-inprocessing downgraded to suggestive; no-eliminate null; push-$n$ campaign $n \in \{70..100\}$ added to v6 plan as obligated continuation.

*End of v5 draft. The paper is offered as a methodology contribution; no complexity-theoretic separation is claimed. All exponents reported are heuristic-conditional and schedule-specific, with no formal ordering to* $\text{CSpace}_{\text{cum}}$ *per v5-corrected Lemma A. The ADRNV open problem on Tseitin (p. 38:19, lines 996–998) is reaffirmed as open.*

## References


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